

D Manish Kumar

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SKILLS SUMMARY

- **Languages:** Python
- **Frameworks:** Pandas, NumPy, Matplotlib
- **Tools/Platforms:** Proteus, AutoCAD

PROJECTS

- **Finger Angle Tracking with Servo Motor** October 2024
Developed a real-time gesture-controlled system that detects the angle between the user's index finger and thumb using computer vision and moves a servo motor accordingly.
- **Gesture-Controlled LED** November 2024
Built a basic system that turns an LED on/off using finger gestures. Utilized computer vision and microcontroller integration for gesture recognition.
- **ESP32 – Contactless Thermometer:** April 2025
Developed a non-contact thermometer capable of accurately measuring body temperature using the MLX90614 IR sensor and ESP32 microcontroller. Leveraged the ESP32's Wi-Fi capabilities for potential IoT integration and future remote data logging or alert systems. Implemented sensor interfacing and data processing using Arduino IDE, ensuring reliable temperature readings suitable for health monitoring applications.
- **RC Vehicle with Assistive Sign Language Translation** March 2025
Designed and developed a remote-controlled (RC) vehicle equipped with robotic "helping hands" powered by servo motors, aimed at assistive communication. Integrated a camera module to capture hand gestures, which were then processed (via external software or systems) to translate sign language into spoken audio, enhancing communication for hearing- or speech-impaired individuals. Focused on hardware implementation using Arduino for motor control and gesture mimicry.
- **Stock Market Analysis & Prediction Tool** March 2025
Created a desktop application for stock analysis and prediction using Python. Integrated the Yahoo Finance API (yfinance) to fetch historical stock data and visualized it with Matplotlib, including candlestick charts using mplfinance. Built an intuitive GUI with Tkinter for user interaction and implemented a predictive model using RandomForestRegressor from scikit-learn to forecast stock prices. Evaluated performance with multiple error metrics (MAPE, MAE, RMSE, R^2), allowing users to analyze trends and get basic price

CERTIFICATIONS

- AutomateX: Mastering PLCs for Industrial Innovation January 2025

EDUCATION

- **Lovely Professional University** Punjab, India
Bachelor of Technology - Electronics and Communication Engineering Since August 2024
- **Vikash The Concept School** Sambalpur, Odisha
Intermediate; Percentage: 62% April 2023 - March 2024
- **Kendriya Vidyalaya** Rourkela, Odisha
Matriculation; Percentage: 70% April 2022 - March 2023